Production Web Engineering & Deployment Architecture Handbo

Category: Website Development • Lead Mentor: Nasir (Lead Technical Mentor) • Licensed for Educational Study

1. Production Web Stack Architecture

- Full-Stack Architecture: Separation of Concerns between Edge Presentation and State Persistence.
- Node.js & Express API Bridge: Unified routing layer capable of serving static assets and dynamic endpoints.
- Serverless vs Long-Running Containers: Balancing cold-start latency against state persistence requirements.
- DNS Configuration: Apex records (A/ALIAS), CNAME delegation, and automated SSL certificate renewal.

2. Core Web Vitals & Performance Engineering

- Largest Contentful Paint (LCP): Preload critical hero images, eliminate render-blocking CSS, and self-host fonts.
- Interaction to Next Paint (INP): Keep main thread tasks under 50ms; defer heavy DOM calculations to microtasks.
- Cumulative Layout Shift (CLS): Explicitly define width and height attributes on all image, video, and ad containers.

3. Production Deployment Checklist

- Strict Content Security Policy (CSP) and HTTPS strict transport security.
- Cache-Control headers: Immutable static assets with content hashing, strong validation on HTML roots.
- Zero-Downtime Deployment strategies using atomic symlinks or blue/green container deployments.